Friction and Lagrangians

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Abstract

Can we have a Lagrangian that describe friction? We attempt to answer this question here. This note is based on the discussions I had with Joshua, Dima (not professor) and Dima.

1 Introduction

Can we have a Lagrangian that describe friction? This question is vague enough to have both yes and no as answers. Let us therefore start with making the question more precise. First, what is friction? Whenever a particle is moving around in the real world, it is interacting with a large number of other particles in the medium around it. These interactions are impossible to keep track of. Instead of giving up, we model this collective mess as one external force in Newton's law. Let us take the oscillator. From experience, we realize that friction opposes the direction of motion, and this prompts us to model friction as a force *opposite to* the direction of motion,

$$\ddot{x} + \gamma\dot{x} + \omega^2 x = 0 \tag{1.1}$$

The second term above is called the friction, and it models our ignorance about the interaction with the medium. We can then immediately ask a question: can we get the friction term $\dot{x}$ from an action principle? The answer is no. To prove this, let us imagine that we can get friction from action. Then,

$$\frac{\delta S[x]}{\delta x(t)} = \dot{x}(t) \underset{\text{discretise}}{\longrightarrow} \quad \frac{\partial S(x_1, \cdots, x_N)}{\partial x_m} = \frac{x_{m+1} - x_m}{\epsilon} \tag{1.2}$$

We can now take a second derivative $\frac{\partial}{\partial x_k}$ and see that the second derivative is not symmetric under $m \leftrightarrow k$. This immediately proves that we cannot get the friction force term from an action principle. Is that the end of story? Well, take the following Lagrangian,

$$L(x, \dot{x}, t) = \frac{1}{2} e^{\gamma t} (\dot{x}^2 - \omega^2 x^2) \tag{1.3}$$

The equations of motion for this Lagrangian is precisely (1.1)[1]. What is going on? Here, the above proof is bypassed by getting,

$$\frac{\delta S[x]}{\delta x(t)} = (\ddot{x}(t) + \dot{x}(t)) e^{\gamma t} \tag{1.4}$$

It is easy to verify that the action corresponding to (1.3) do not violate the symmetry of the second variation. The first proof only prohibited getting lone $\dot{x}$ from the action.

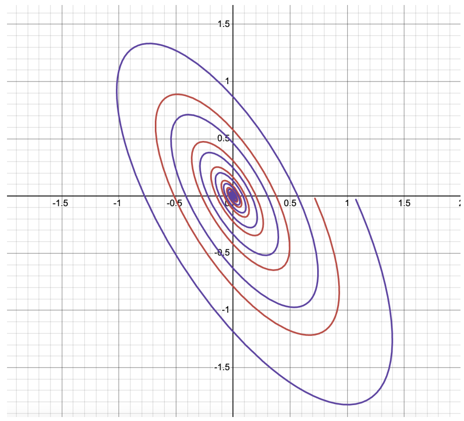
The usual wisdom is that one cannot get friction from a Lagrangian, and yet we have provided a Lagrangian that does it. This is the mystery that we will try to sort out through these notes.

2 Lioville's Theorem

The friction force is not time reversal invariant. That is, if we reverse the direction of time, the equation of motion (1.1) changes. This tallies with our daily experience. There are multiple ways of rolling a ball on the floor such that it comes to a stop at the same position. Given the final state (ball with zero velocity at a corner of the room), we cannot calculate backwards and figure out where it came from, simply because there are (infinitely) many starting conditions that can end up at the same final state. The information about the initial conditions is lost. This is a trade mark of friction, and this is made concrete through a result known as the *Lioville's theorem*.

In the classical world, the state of a particle is given by its position, x and momentum, $p = \dot{x}$. As a particle move, it traces out a path in the $x - p$ space, called the *phase space*. What would the phase space look like for a particle with friction? Let us again take the oscillator. As we mentioned before, there are infinitely many initial conditions that will end up at the same final state, namely oscillator resting at the equilibrium. That means any point in the phase space of the damped oscillator will end up at the origin, as shown below,

[1] There is an overall $e^{\gamma t}$ that multiply (1.1), but this factor is never zero.



We see that if we start with a box of points in phase space, that box will shrink in size until it becomes zero. That is, the volume of a block in phase space decreases over time. Let us make this notion concrete. Consider an infinitesimal volume in phase space at time t,

$$\Delta V(t) = dx(t) \wedge dp(t) \tag{2.1}$$

If we let time increase by a small step ϵ, the position and momentum become,

$$x(t+\epsilon) = x(t) + \epsilon \dot{x}(t) + \mathcal{O}\big(\epsilon^2\big) \quad \text{and} \quad p(t+\epsilon) = p(t) + \epsilon \dot{p}(t) + \mathcal{O}\big(\epsilon^2\big) \tag{2.2}$$

The volume after the time step is then,

$$\Delta V(t+\epsilon) = dx(t+\epsilon) \wedge dp(t+\epsilon) \tag{2.3}$$

Now,

$$\begin{aligned} dx(t+\epsilon) &= \frac{\partial x(t+\epsilon)}{\partial x(t)} dx(t) + \frac{\partial x(t+\epsilon)}{\partial p(t)} dp(t), \\ dp(t+\epsilon) &= \frac{\partial p(t+\epsilon)}{\partial x(t)} dx(t) + \frac{\partial p(t+\epsilon)}{\partial p(t)} dp(t) \end{aligned} \tag{2.4}$$

plugging in give us,

$$\Delta V(t+\epsilon) = (\det J) dx(t) \wedge dp(t) = (\det J) \Delta V(t) \tag{2.5}$$

where J is called the Jacobian,

$$J = \begin{pmatrix} \frac{\partial x(t+\epsilon)}{\partial x(t)} & \frac{\partial x(t+\epsilon)}{\partial p(t)} \\ \frac{\partial p(t+\epsilon)}{\partial x(t)} & \frac{\partial p(t+\epsilon)}{\partial p(t)} \end{pmatrix} = \begin{pmatrix} 1 + \epsilon \frac{\partial \dot{x}}{\partial x} & \epsilon \frac{\partial \dot{x}}{\partial p} \\ \epsilon \frac{\partial \dot{p}}{\partial x} & 1 + \epsilon \frac{\partial \dot{p}}{\partial p} \end{pmatrix} \tag{2.6}$$

The determinant of the Jacobian is,

$$\det J = 1 + \epsilon \left(\frac{\partial \dot{x}}{\partial x} + \frac{\partial \dot{p}}{\partial p} \right) + \mathcal{O}\big(\epsilon^2\big) \tag{2.7}$$

If the $\mathcal{O}(\epsilon)$ term above vanish, the volume of phase space do not change as time evolves infinitesimally, and hence by induction for any arbitrary time step. In the friction problem, we have,

$$\begin{aligned} \dot{x} &= p, \\ \dot{p} &= -\omega^2 x - \gamma p \end{aligned} \tag{2.8}$$

This immediately tells us,

$$\det J = 1 - \gamma\epsilon + \mathcal{O}\big(\epsilon^2\big) \tag{2.9}$$

The determinant is decreasing, telling us that the volume is decreasing over time, confirming our intuition.

Let us recall that if the system is described by a Hamiltonian, then phase space volume will never shrink. This is easy to see. For a Hamiltonian system, we have,

$$\dot{x} = \frac{\partial H}{\partial p} \quad \dot{p} = -\frac{\partial H}{\partial x} \tag{2.10}$$

Using these, we see that the $\mathcal{O}(\epsilon)$ term in the determinant will vanish due to the symmetry of the partial derivative and the all important minus sign in Hamilton's equations. Notice that here we did not assume anything about the time dependence of the Hamiltonian, all we needed was to assume that it exists.

Is there any way to have the volume be non-shrinking in the friction system? There is a cheap trick that we can employ. Noticing that the decreasing of the phase space volume is solely due to the decrease of p, we do the following: At each time step, we re-scale the p axis to counter act the shrinkage in volume, such that the volume in the new rescaled phase space is the same as the old (non-rescaled) volume that we started with. We repeat this at each time step. After a bit of trial and error, we find the following re-scaling,

$$\tilde{p}(t) = e^{\gamma t} p(t) \tag{2.11}$$

The equations of motion become,

$$\begin{aligned} \dot{x} &= e^{-\gamma t}\tilde{p}, \\ \dot{\tilde{p}} &= -\omega^2 e^{\gamma t} x \end{aligned} \tag{2.12}$$

We immediately see that in the $x - \tilde{p}$ space, the volume of a box do not shrink under the dynamics. We can even find a Hamiltonian by staring the above equations,

$$H = \frac{1}{2} e^{-\gamma t}\tilde{p}^2 + \frac{1}{2}\omega^2 e^{\gamma t} x^2 \tag{2.13}$$

Upon Legendre transformation, we find that the Lagrangian corresponding to the Hamiltonian is precisely (1.3)!

So can we say that the Lagrangian (1.3) has no dissipation at all? Given just the Lagrangian, we will be naturally looking at the phase space defined by the canonical position and momentum, and in that phase space, volumes are conserved, implying no dissipation. However, we cannot compare this phase space behavior with that of the phase space for (1.1), precisely because momenta in the two systems are defined differently. If we insist to look at the phase space defined by x and $\dot{x}$ for the Lagrangian system, we will indeed see a reduction of volume. This is not a contradiction to Lioville's theorem. Lioville theorem only tells us that the canonical phase space volume is conserved for a Hamiltonian (or Lagrangian) system.

3 Conclusions

What does this all mean? Let us summarize the results through a series of questions and answers.

Q : Can we get friction from Lagrangian?
A : Yes, provided that we declare x and $\dot{x}$ are the observables. However, the Lagrangian defines the canonical variables. If we are not going to use the canonical variables, why bother writing a Lagrangian?

Q : But the existence of a Lagrangian imply volume of phase space is conserved, but friction should not do that?
A : The volume of the canonical phase space is conserved. By declaring that x and $\dot{x}$ are the physically observable variables, one should look at the phase space defined by these two and not the canonical x and $\tilde{p}$ phase space. In the $x - \dot{x}$ space, volume is not conserved.

Q : The whole Lioville's theorem business seem to rely on how we define our phase space. Should we then trust this theorem at all?
A : To draw meaningful conclusions from Lioville's theorem, one should look at the phase space defined by physically observable variables. Looking at phase space defined by some random position and canonical momentum can mislead you.

Q : So can we get friction from a Lagrangian such that the canonical variables are the physical position and momentum?
A : No. Lioville's theorem (in the $x - \dot{x}$ space) prohibits that.

Q : Can we find a different set of phase space variables such that effect of friction is nullified in general for any dissipative system?
A : I don't know! Does not seem to be true, but that is just a guess.

Q : In the linear friction case that we discussed above, is there any alternate way to "get rid of" friction?
A : There is an equivalent way, by re-scaling time. The particle slows down in our normal time coordinate, so just slow down rate of change of time itself, so that the particles velocity stays constant (something like that). This can be done by defining $\tau(t)$ such that $dt\ e^{-\gamma t} = d\tau$. The Lagrangian then becomes,

$$S = \frac{1}{2}\int dt\ e^{-\gamma t}\big((\dot{x}e^{\gamma t})^2 - e^{2\gamma t}\omega^2 x^2\big) = \frac{1}{2}\int d\tau\left(\left(\frac{dx(\tau)}{d\tau}\right)^2 - \frac{1}{\gamma^2\tau^2}\omega^2 x^2\right)$$

We can also try the same substitution in the equation of motion (1.1), and see that it indeed cancels the friction term. This is exactly the same as re-scaling momentum, as we can see from the definition of canonical momentum here.